Stocktrails

FINTECHATHON 2026 · INTERNATIONAL TRACK

Technical documentation

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Summary

Stocktrails is a Chrome browser extension. It records which stocks an investor looks at on finance portals, when, for how long and on which kind of page, together with the calls they make (Positive, Neutral or Negative, with an optional target, horizon and note). Investors can import their broker's transaction export, which is read and stored only on their device.

On top of these two records sits a behavioural finance agent. It **recognises** the investor's own biases with statistical tests, **recommends a concrete rule** for each only after checking that the rule would have improved later trades it had not seen, lets the investor **adopt** the rule with one click, then **tracks adherence** and checks intended buys against adopted rules. An assistant answers questions about the investor's data by choosing read-only tools that run on the device.

The agent also **measures the value it adds**. Next to the investor's actual portfolio it keeps a paper portfolio that makes the same buys under the rules the investor adopted. The gap between the two, per buy and with a confidence range, is what keeping one's own rules is worth in returns. The investor sees the benefit in their own numbers, not as a promise.

Statistics decide

Every verdict comes from deterministic, tested code. The language model chooses tools and words the result.

Trades stay local

Imported trades, amounts and identifiers never leave the device. Three filters guard what does.

Out of sample first

A rule is recommended only if it held up on later decisions. In 24 worlds with no effect, 0 of 216 rules passed.

Value you can measure

Your portfolio against the same buys made under your adopted rules, per buy, with 99% ranges.

No order path

No tool, endpoint or credential exists that could place a trade. Every change is a click.

1 System architecture

The product is local-first. The client is a Chrome Manifest V3 extension: a content script on finance portals, a service worker, and a dashboard page that holds all analysis. A Supabase project in the EU provides sign-in, synchronisation of attention data between a user's browsers, anonymous community aggregates and a small set of edge functions. The heavy analysis runs in the browser.

1.1 Components and trust zones

Figure 1 groups the components by where they run. The device zone holds all raw data and all analytics. The cloud zone holds only attention events that the user chose to sync and per-user counters. Third-party services receive price requests, server-side data collection requests, or de-identified results for the model.

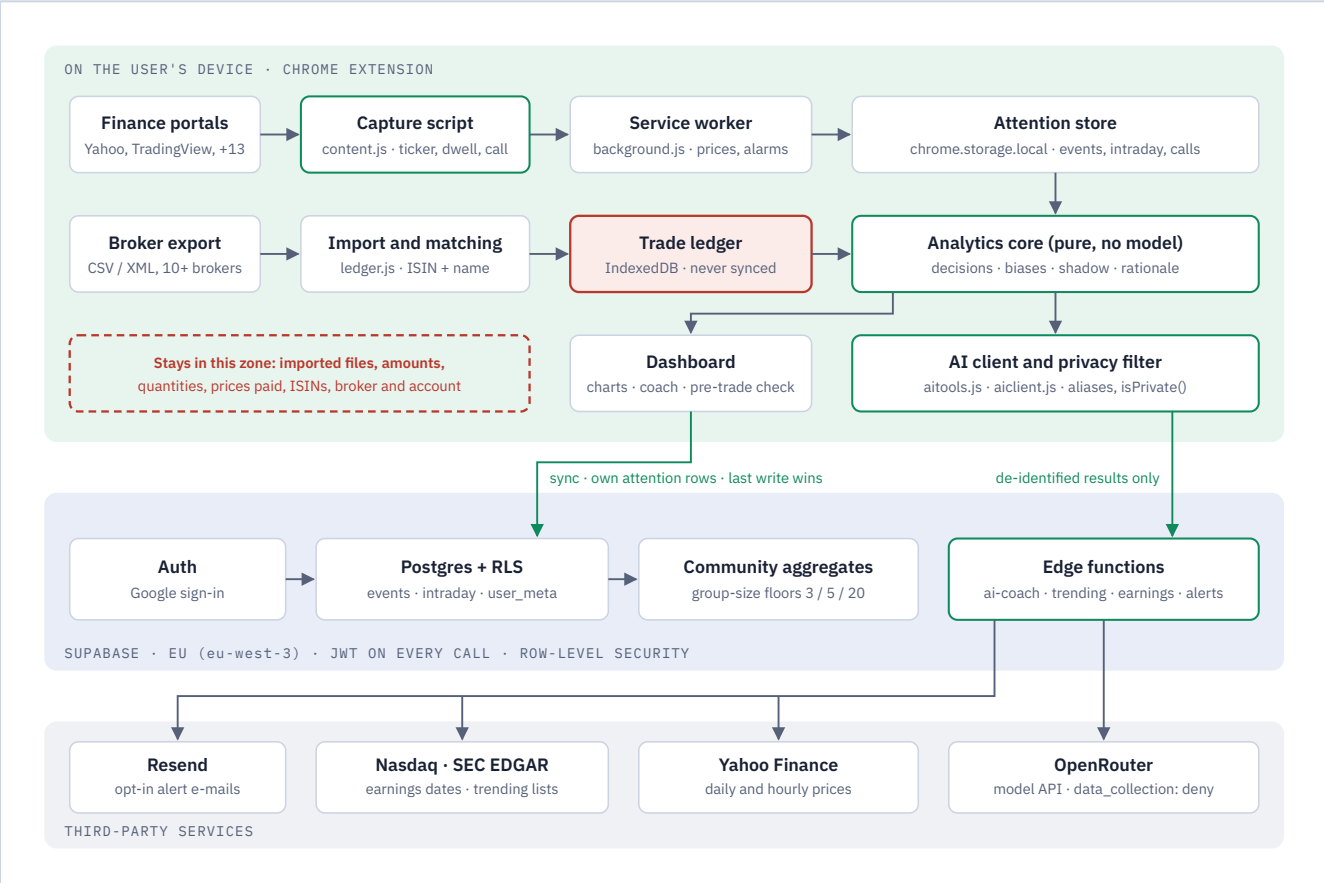


Figure 1. Components and trust zones. Green arrows are the only two paths by which user data leaves the device. The extension fetches prices from Yahoo directly; only the ticker and the date range are sent.

COMPONENT	CODE	ROLE
Capture script	`src/content.js`, `src/portals.js`	Detects the ticker from the URL (15 built-in portals and user-defined patterns), falls back to page content (cashtags, exchange-qualified symbols, tracked names). Measures active visible time. Renders the call card in a closed shadow root.
Service worker	`src/background.js`	Message router, price at the moment of the view, hourly price archive, alarms (sync every 5 min, daily prices), viewing sessions for the AI agent's live verdict.
Local stores	`store.js`, `ledgerdb.js`, `pricedb.js`, `pricecache.js`	Attention events and settings in `chrome.storage.local`; trade ledger, adopted rules, agent decisions and pre-trade checks in IndexedDB; price archives in IndexedDB.
Analytics core	`decisions.js`, `biases.js`, `shadow.js`, `rationale.js`, `indicators.js`	Pure functions, no I/O and no model: decisions and outcomes, bias tests, rule fitting and walk-forward validation, adherence, paper portfolios, statistics on notes.
AI client	`aitools.js`, `aiclient.js`	Read-only tool catalogue built from allowlists, aliases, privacy check, the agent loop, number checking, action cards and the request log.
Cloud	`supabase/`	Postgres with row-level security, Google sign-in, community RPCs with group-size floors, edge functions (`ai-coach`, `trending`, `earnings`, `crowd-alerts` and helpers).

1.2 Where each kind of data lives

DATA	STORED	LEAVES THE DEVICE	DELETED BY
Attention events (ticker, time, portal, page section, dwell, price at view, call)	Browser; Supabase after sign-in	`own account only`	Delete all, or account deletion
Imported broker trades, matches	IndexedDB `stocktrails_ledger`, per account	`never`	Delete all; not in JSON backup

DATA	STORED	LEAVES THE DEVICE	DELETED BY
Adopted rules, reminders, review notes	IndexedDB	`never`	Delete all
Paper decisions of the agents, pre-trade checks	IndexedDB stores `agent`, `checks`	`never`	Delete all
Tool results and questions sent to the assistant	Local request log (last 80 entries)	`de-identified, on consent`	Clear log; sign-out
Assistant usage	Supabase `ai_usage`: user id, day, count	counter only	Account deletion
Community aggregates	Computed at query time from synced events	only above group-size floors	n/a

1.3 Capture and sync

Capture needs no click. The content script polls the URL every 1.5 seconds; a new ticker becomes an attention event. Each event is written twice: to the *daily* set, where repeat views within 60 minutes collapse into one row with a view counter, and to the *intraday* set, which keeps every view. All writes land locally first, so the dashboard works offline.

Synchronisation between a user's browsers uses two cursors because push and pull ask different questions. The push cursor compares local timestamps with local timestamps. The pull cursor uses the server's own receive time (`synced_at`), because a row captured early but uploaded late would otherwise be invisible to devices whose cursor had moved past its capture time. Pulls page by keyset in blocks of 1000 rows, since PostgREST truncates responses at that size and offset paging skipped rows that changed during a pull.

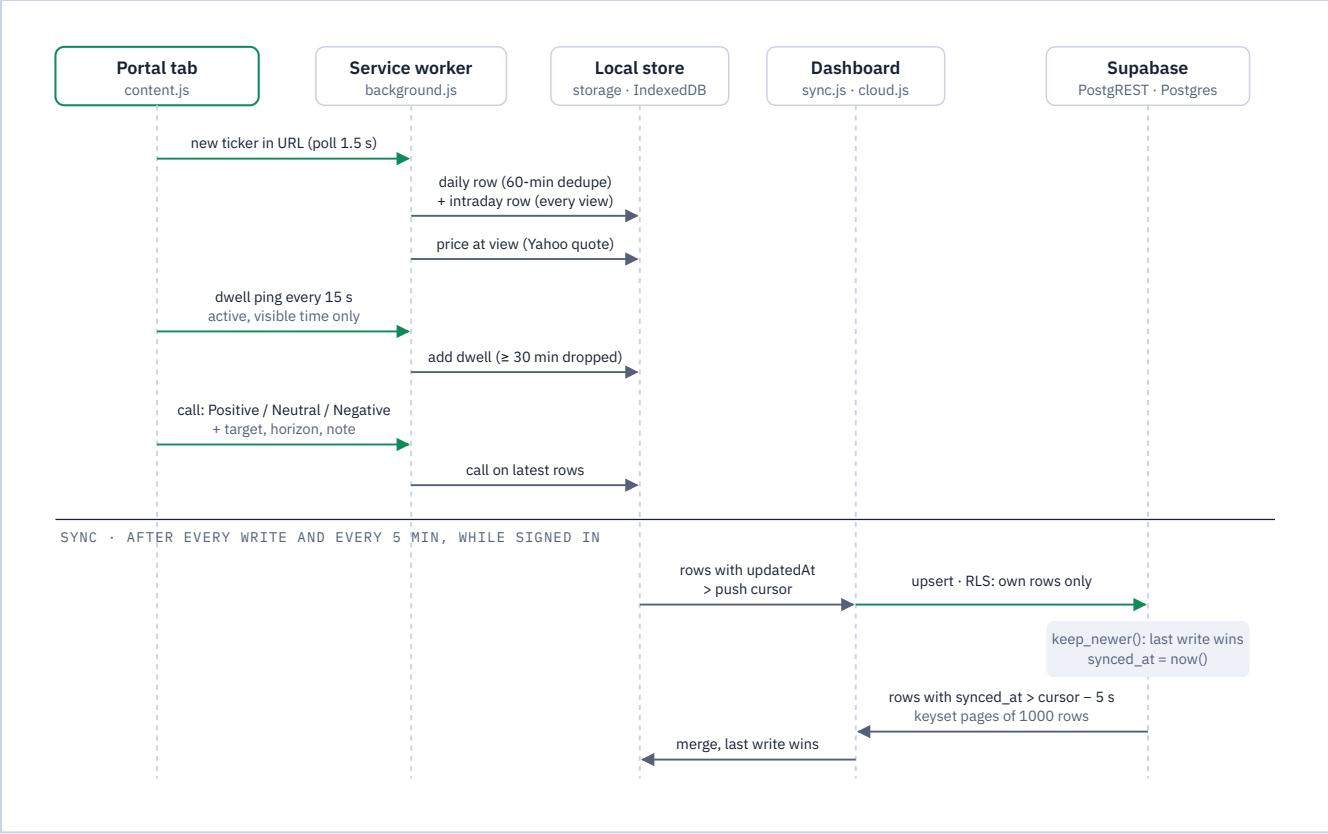


Figure 2. Capture and sync for one page view. A full reconcile that ignores both cursors runs once a day and whenever the sync protocol version changes, so any gap closes by itself. The trade ledger is not part of sync.

1.4 Technology and deployment

LAYER	TECHNOLOGY	DELIVERY
Extension	Chrome Manifest V3, plain JavaScript, no remote code, no build step	Chrome Web Store, published from GitHub Actions
Backend	Supabase: Postgres, Auth, RLS, pg_cron, Vault, Deno edge functions	SQL migrations; functions deployed with JWT verification on
Model	OpenRouter (OpenAI-compatible chat API, plus its Decisions API for the AI agent's choices); every request carries `data_collection: "deny"`; a stand-in model for offline demos	Called only from the `ai-coach` function; one module builds and reads requests, so the model and provider are replaceable

2 Core algorithms

The central design decision is the division of labour. Everything that produces a number or a verdict is deterministic JavaScript, tested without a model. The model is used for four things only: choosing which read-only tools to run for a question, putting the returned results into words, proposing actions the user may approve, and, as one of two paper-trading agents, making decisions that are compared with the user's and the rules agent's.

2.1 Pipeline

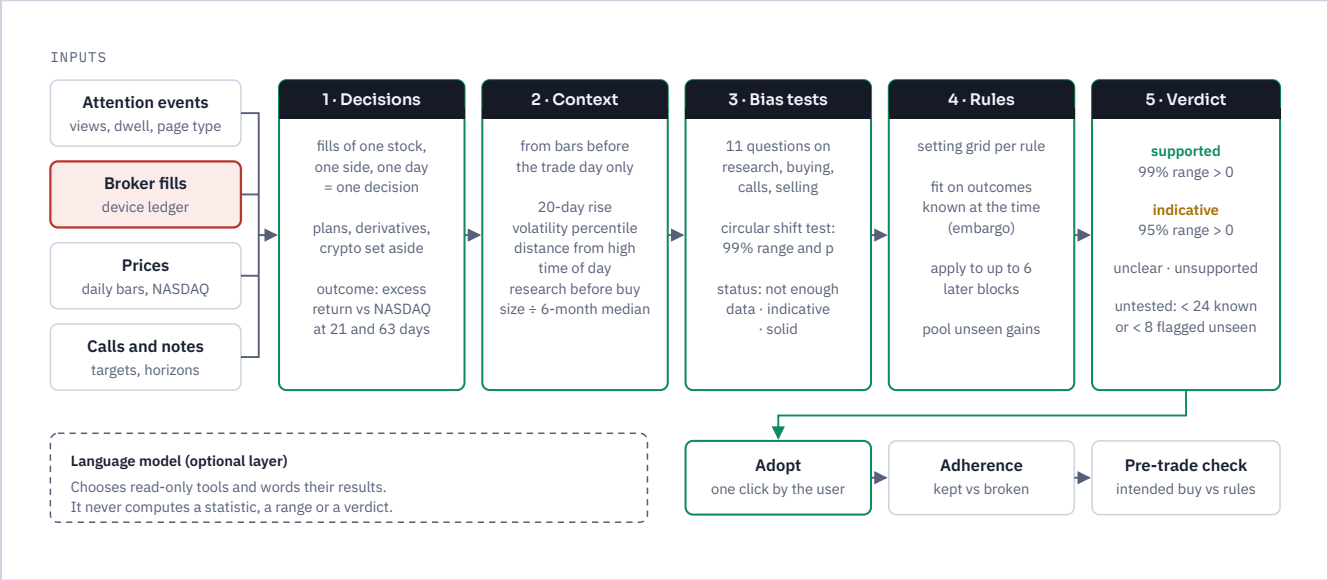


Figure 3. From raw records to a recommendation. Steps 1 to 5 are pure functions in `src/decisions.js` and `src/biases.js`. A rule reaches the user only from step 5, and changes behaviour only after the user adopts it.

2.2 Biases tracked and their sources

Each bias is measured from the investor's own records with a stated sample-size status (not enough data, indicative, solid). The measures adapt definitions from the literature to what a browser can observe: research behaviour, calls and imported trades. Full references are listed at the end of the document.

BIAS	WHAT THE APP MEASURES	LITERATURE
RESEARCH AND ATTENTION		
Attention-driven buying	Buys that follow a spike in your own views, or in other users' views	Barber & Odean 2008; Da, Engelberg & Gao 2011

BIAS	WHAT THE APP MEASURES	LITERATURE
Buying without research, impulse buying	Time from first look to buy; buys with no recorded research	Barber & Odean 2008; Kahneman 2011
Ostrich effect	Fewer looks at held stocks after market falls	Karlsson, Loewenstein & Seppi 2009; Sicherman et al. 2016
Confirmation bias	Calls not revised after contrary news; narrow mix of page types read	Nickerson 1998; Park et al. 2013
Herding	Share of first looks that follow other users' attention	Kumar & Lee 2006; Barber, Odean & Zhu 2009
Limited attention	Delay before looking at a tracked or held stock after its earnings release	DellaVigna & Pollet 2009; Hirshleifer, Lim & Teoh 2009
BUYING		
Chasing price rises, extrapolation	Buys after a run-up or a strong market stretch; calls against past 1, 3 and 12-month returns	De Bondt 1993; Benartzi 2001; Greenwood & Shleifer 2014
Lottery stocks	Share and size of buys in stocks with lottery features (low price, high volatility and skew)	Kumar 2009; Bali, Cakici & Whitelaw 2011
Volume-driven buying	Buys after abnormal trading volume	Gervais, Kaniel & Mingelgrin 2001; Barber & Odean 2008
Buying at a bad time of day	Outcomes of buys by time of day, robust to moved boundaries	Danziger, Levav & Avnaim-Pesso 2011; Hirshleifer et al. 2019
Oversized buys in turbulent markets	Size against your usual buy, by market volatility percentile	Hoffmann, Post & Pennings 2013
Timing instead of a plan	Your timed buys against the same money invested monthly	Dichev 2007
YOUR CALLS		
Not acting on your own calls	Positive calls not bought, and Negative calls on held stocks not sold, within 30 days	Sheeran 2002

BIAS	WHAT THE APP MEASURES	LITERATURE
Anchoring	Round targets; targets near the 52-week high or entry price; sales at break-even	Tversky & Kahneman 1974; George & Hwang 2004; Li & Yu 2012
Overconfident calls	Required move against the best move before the deadline; Brier score	Lichtenstein, Fischhoff & Phillips 1982; Brier 1950
SELLING AND HOLDING		
Disposition effect	Winners sold sooner than losers (holding-period survival)	Shefrin & Statman 1985; Odean 1998
Rank effect	Probability of selling your best and worst positions	Hartzmark 2015
Myopic loss aversion	Checking frequency and sales after dips in long-horizon positions	Benartzi & Thaler 1995; Gneezy & Potters 1997
Repurchase regret	Rebuying stocks sold at a loss against those sold at a gain	Strahilevitz, Odean & Barber 2011
PORTFOLIO AND ACTIVITY		
Overtrading	Turnover; returns of switches from one stock to another	Odean 1999; Barber & Odean 2000, 2001
Under-diversification	Number of positions; concentration by value, sector and country	Benartzi & Thaler 2001; Goetzmann & Kumar 2008
Home bias	Home share of views and holdings against the country's world-index weight	French & Poterba 1991; Huberman 2001; Grinblatt & Keloharju 2001

Table 1. Biases the app tracks, grouped by decision stage.

2.3 Decisions and outcomes

Fills of one instrument on one side on one day form a **decision**. Savings-plan executions, derivatives, bonds and crypto are set aside, as are decisions made before any research was recorded. The outcome of a buy is its excess return over the NASDAQ Composite (`^IXIC`) from the close of the trade day, over 21 trading days, and over 63 days for calls. Holdings that existed before the export begins are inferred from sales larger than the known position.

Context is measured from bars *before* the trade day, so nothing the investor could not have seen enters a feature. Volatility is the standard deviation of 20 daily returns; the buy is then placed by percentile among

all trading days of the covered period, so that "you buy in turbulent markets" is judged against how often markets are turbulent. Size is judged against the investor's median buy over the previous 6 months, so a growing portfolio does not read as buying bigger.

2.4 The circular shift test

Markets stay calm or turbulent for weeks, and buying comes in bursts. Neighbouring days and neighbouring buys are therefore alike, and an ordinary bootstrap gives ranges that are too narrow. Instead, the whole pattern of buys is slid along the calendar by every possible offset, wrapping around at the end, and each statistic is recomputed. The buys keep their spacing and the market keeps its stretches; only their alignment is broken. The distribution over all offsets gives the 99% range and the p-value of the observed statistic (`shiftTest`).

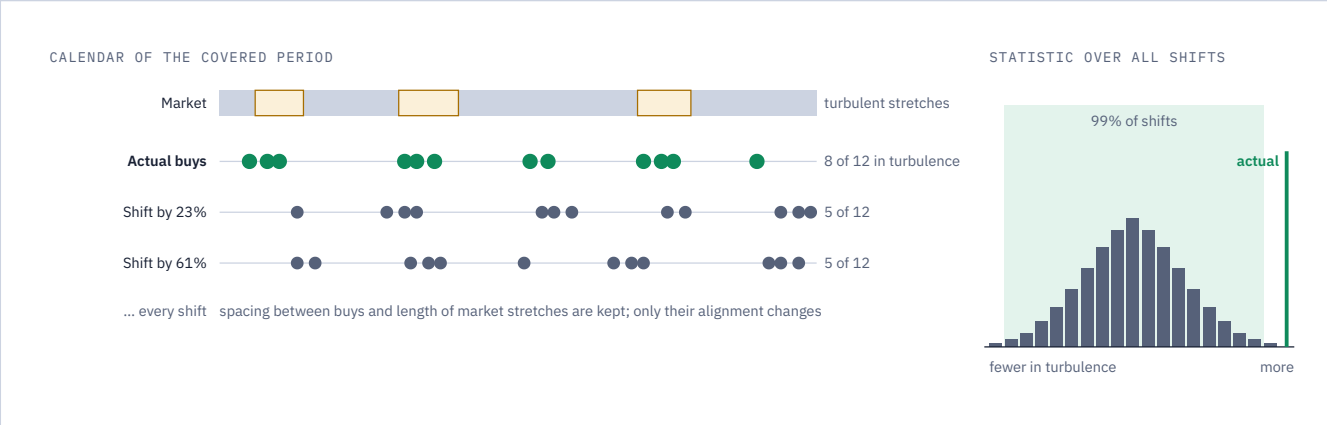


Figure 4. The shift test on an illustrative pattern of 12 buys. If the actual share of buys in turbulent markets lies outside the range produced by all shifts, the alignment of buying with turbulence is unlikely to be chance.

2.5 Rules and walk-forward validation

Each bias maps to one or more rules, for example "wait after a run-up", "sleep on it", "avoid your worst time of day" or "cap the size in turbulent markets". A rule has a grid of settings, a *flag* (the rule says do not) and a *gain*: what not doing it would have been worth against the NASDAQ, assuming the skipped money earns the market. A cap rule scales the size instead of skipping.

`fit` chooses the setting that maximises

```
score(s) = mean_gain(s) × √ n_flagged(s)

subject to  n_flagged(s) ≥ 5,  n_flagged(s) ≤ 0.6 × n,  mean_gain(s) >
0
```

`walkForward` orders decisions in time. The first half is only ever used for training. The rest is cut into up to six blocks. For each block, the rule is fitted on decisions whose outcome was **already known when the**

block began: a buy's outcome takes a month to arrive, so the decisions of the month before the block are embargoed. The fitted setting is then applied to the block. A fold in which no setting helped recommends nothing for that block. Only the gains on the later blocks are pooled and given a bootstrap range.

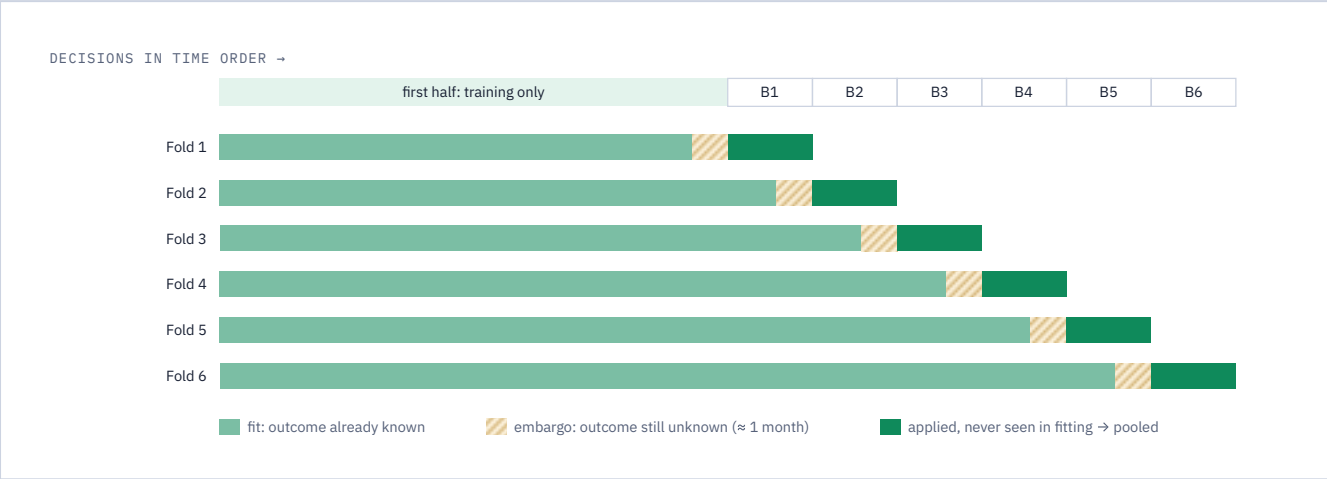


Figure 5. Chronological walk-forward. Each block is judged only by a rule fitted on decisions that had finished by the time the block began. The pooled gains from the six test blocks give the verdict in Table 2.

VERDICT	CONDITION	WHAT THE USER SEES
supported	99% bootstrap range above zero	Recommended, with setting, gains and ranges
indicative	95% range above zero	Shown as a lead, not recommended
unclear	range includes zero	Not recommended
unsupported	range below zero, or in-sample only	Not recommended; the in-sample flattery is shown
untested	fewer than 24 decisions with a known outcome, or fewer than 8 flagged out of sample	"Not enough data", a result in itself

Table 2. Verdict on the pooled out-of-sample gain per flagged decision.

2.6 Adherence and the pre-trade check

After adoption, the coach reports the share of decisions that broke the rule against the same share over an equally long stretch before adoption, with a Newcombe interval for the change, and compares the outcome of decisions that kept the rule with those that broke it, with a bootstrap range. It is reported as an association, since nobody was assigned to comply. The pre-trade check takes an intended buy (stock and amount), measures the same features as of now, and lists which adopted rules object, which cannot

be checked, and any reminder the user set for that stock. Every check is saved on the device with its inputs and the rule versions in force, so checks can be evaluated later.

2.7 The assistant agent loop

The assistant works in steps, at most five requests per question. Each request carries the question, with stocks replaced by aliases, and the results gathered so far. The model either names up to four more tools from a fixed read-only catalogue, or answers. The tools run in the browser, so the model can look further when a first result raises a question, for example by reading a rule's details after seeing the biases, or trying a rule with another setting. The fifth step must answer, and a tool call it already has is dropped. The browser then checks every number in the reply against the results and marks any it cannot find.

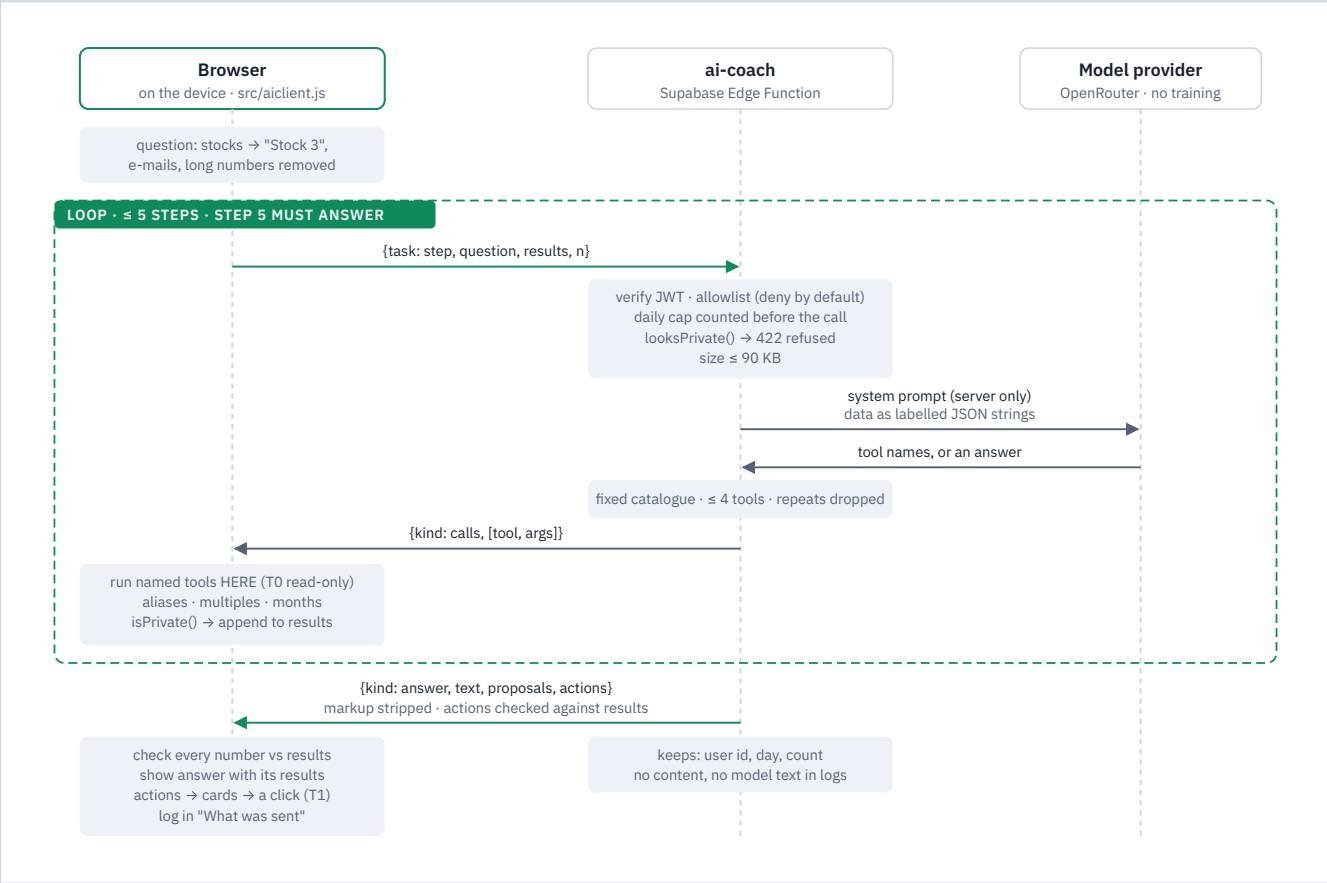


Figure 6. One question through the assistant. Raw data never crosses the dashed lifeline of the device; only tool results that passed the privacy check do. Every step counts against the user's daily cap.

TOOL (T0, READ-ONLY)	RETURNS
`get_overview`, `get_finding(q1…q11)`	Counts, shares, returns against the NASDAQ and statuses, as plain facts
`get_biases`, `get_rule(id)`, `simulate_rule(id)`	Biases, rule wording, the out-of-sample verdict

TOOL (T0, READ-ONLY)	RETURNS
`list_decisions`	Alias, month, size as a multiple of the usual buy, research, rise, time of day, market, result; no amounts
`get_adherence`, `check_buy(stock, size)`	How rules are kept; an intended buy against adopted rules and reminders
`compare_periods(days)`, `what_if(id, …)`	Whether a habit is fading; a rule with another setting over the whole history and its later half
`explain_decision`, `stock_summary`, `get_weekly_review`	One decision with its research and the rules that flag it; one stock's views, calls and decisions; the weekly review
`get_agent_portfolio`, `get_agent_decisions`	The paper portfolios as ratios; what each agent decided at single moments and why

2.8 Two agents with paper portfolios

To show what following the rules would have meant in money terms, two agents trade paper portfolios on the investor's own buys and on the stocks they looked at. The **rules agent** (`src/shadow.js`, no model) keeps, each month, only the rules that had held up out of sample on outcomes known by then, plus any rules the investor adopted. For each candidate it buys at once, waits until no active rule objects (re-checking 1 to 10 trading days later), or passes. The **AI agent** receives each candidate as of that day (alias, "day N", ratios, the rules agent's view) in requests of 15, and is asked only about moments at most two days old. Every decision is saved once with its time; there is no backfill, so the model cannot know what happened next.

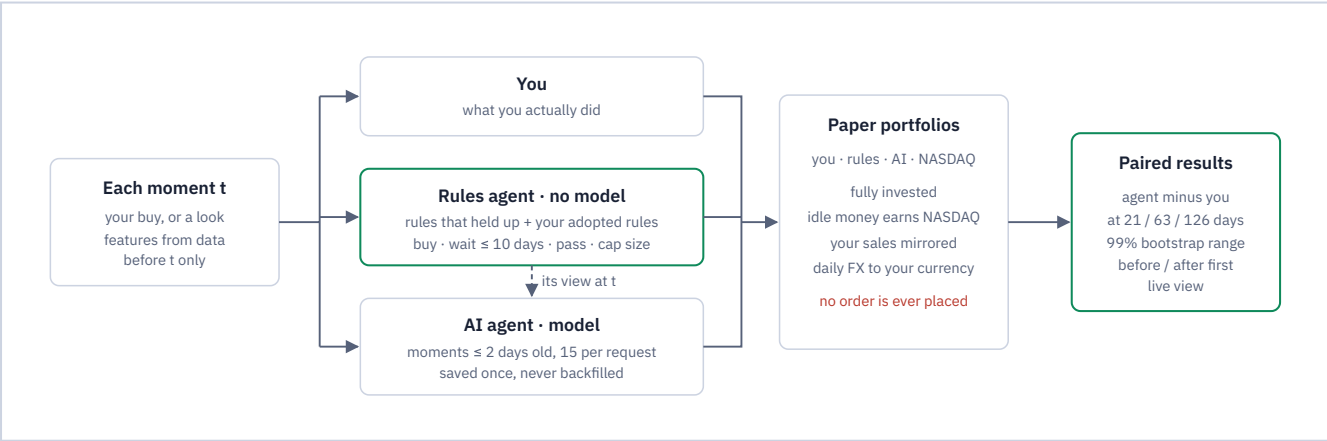


Figure 7. The agent simulation. Both agents act only on the investor's own buys and looks and only move paper portfolios drawn in the browser. Results are measured to the same end bar for each pair.

2.9 Measured value added

Adherence tracking answers the question an investor cares about: do my own rules make me money? From the first imported buy, the app keeps three portfolios side by side on the same money, valued daily in the investor's currency:

- the investor's **actual** portfolio;
- the same buys made **under the investor's adopted rules**: the rules agent with every adopted rule applied, so it waits or passes where a rule objects and cuts the size where a cap applies;
- the **NASDAQ** as a benchmark.

The gap between the first two is the value of keeping one's own rules. It is reported per buy at 21, 63 and 126 trading days, with a 99% bootstrap range, and separately for the period before and after the investor started using the app. The adherence view adds the direct comparison: decisions that kept an adopted rule against those that broke it. The investor therefore sees the benefit of the agent in their own returns, as a measured difference rather than a forecast.

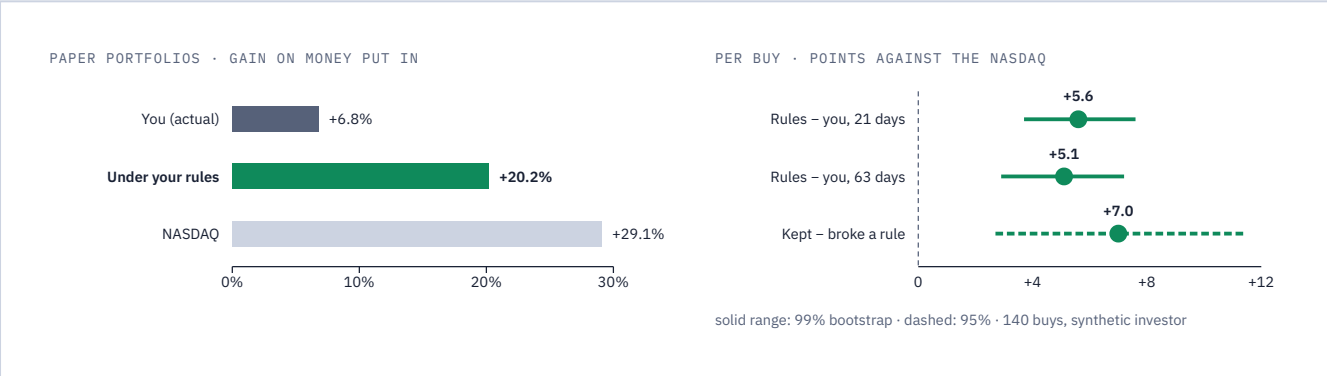


Figure 8. Value added on the synthetic investor of Table 3. Keeping the adopted rules is worth about 5 points per buy against the index, with a range well clear of zero. The index itself still did better over this history; the claim is a measured improvement on the investor's own behaviour, not a promise to beat the market.

2.10 Calibration results

A generator builds deterministic synthetic investors with planted behaviours. The coach must find what is planted, recommend only what holds up on later trades, and find nothing where nothing exists. Table 3 comes from one such investor (seed 21, 140 analysed buys over about 31 months).

RULE	VERDICT	FLAGGED (LATER)	OUT OF SAMPLE	99% RANGE	IN SAMPLE
Wait after a run-up	supported	15 / 70	+16.3	+10.8 to +21.5	+14.2
Avoid your worst time of day	supported	19 / 70	+17.0	+12.3 to +21.1	+15.0

RULE	VERDICT	FLAGGED (LATER)	OUT OF SAMPLE	99% RANGE	IN SAMPLE
Sleep on it (impulse buys)	supported	27 / 62	+13.2	+9.2 to +17.2	+12.1
Research before you buy	supported	41 / 70	+11.5	+7.9 to +15.1	+10.0
Act on your calls or drop them	supported	53 / 89	+11.3	+6.5 to +16.0	+10.9
Pause in turbulent markets	supported	26 / 70	+10.9	+6.3 to +15.1	+10.9
Cap the size in turbulent markets	supported	18 / 68	+4.4	+2.6 to +6.2	+3.9

Table 3. Rules on a synthetic investor. Gains are percentage points against the NASDAQ per flagged decision, out of sample, with 99% ranges.

> **Null worlds.** In 24 synthetic worlds with no planted effect, 0 of 216 rules were called supported. An effect that stops halfway through the history is not recommended, and the embargo has its own unit tests.

3 Security design

The security design follows from one principle: the most sensitive data, an investor's trades, should not exist anywhere it does not need to. Raw trades are parsed in the browser and kept in IndexedDB on the device. The cloud receives attention events for the user's own account under row-level security, and the model receives only de-identified results that pass three filters. The agent has no path to an order.

Trust boundaries, in order of exposure: (1) finance portal pages, which are untrusted and may read the DOM around the capture card; (2) the device, which holds everything; (3) the Supabase project, which holds synced attention and counters; (4) model and data providers, which see only what the edge functions send.

3.1 Data minimisation

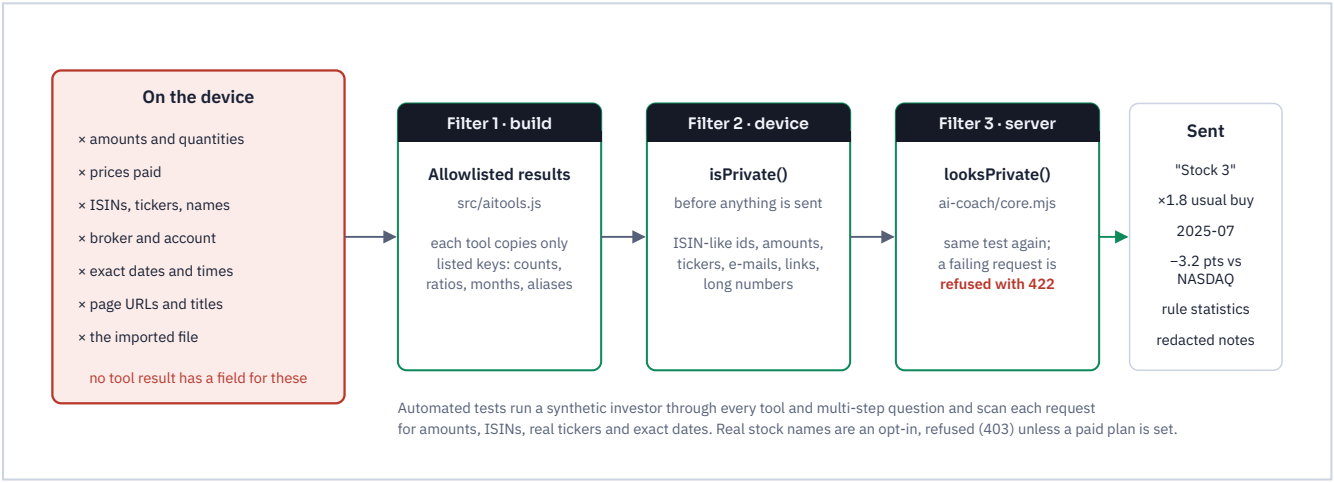


Figure 9. Defence in depth for everything sent to the model. Filter 1 makes leakage structurally hard; filters 2 and 3 catch what a future code change might let through.

3.2 Permission tiers

The agent's authority is fixed in code. The server validates every tool call and proposed action, and the browser executes only what the user confirms.

	What the agent may do	How it is enforced
T0 READ-ONLY	Name tools from a fixed catalogue that read results computed on the device. As the AI agent: say buy, wait 1 to 10 trading days or pass on a paper portfolio.	Server accepts only `TOOLS` in `core.mjs`, checks every argument, ≤ 4 tools per step, ≤ 5 steps. Paper decisions: asked ids only, three actions, clamped days, ≤ 15 per request.
T1 USER DECIDES	Propose an action: adopt, retire or retune a rule, set a pre-trade reminder for a stock, save a review note.	Validated against `ACTIONS`: known types, rule ids and stocks that appear in the results, at most 3. Shown as a card with what-if numbers; written locally only on a click, logged as "You approved".
T2 NOT OFFERED	**What the agent can never do**: Place or change orders, hold broker or bank credentials, write trade data to any server, change account settings.	No tool, endpoint, scope or code path exists. The function has no database access beyond upserting a usage counter.

Figure 10. Permission tiers as implemented. Each tier is covered by automated tests.

3.3 Cloud authorisation

Row-level security is enabled on every table. Owner policies use `user_id = auth.uid()` for reading and writing. Admin reads go through `is_admin()`, which a user cannot set because `profiles` has a column-level UPDATE grant. Service-only tables (`ai_usage`, `crowd_notified`) have no policies, and their functions are revoked from `anon` and `authenticated`. Security-definer functions pin `search_path`. Community views return a cell only when enough distinct users stand behind it.

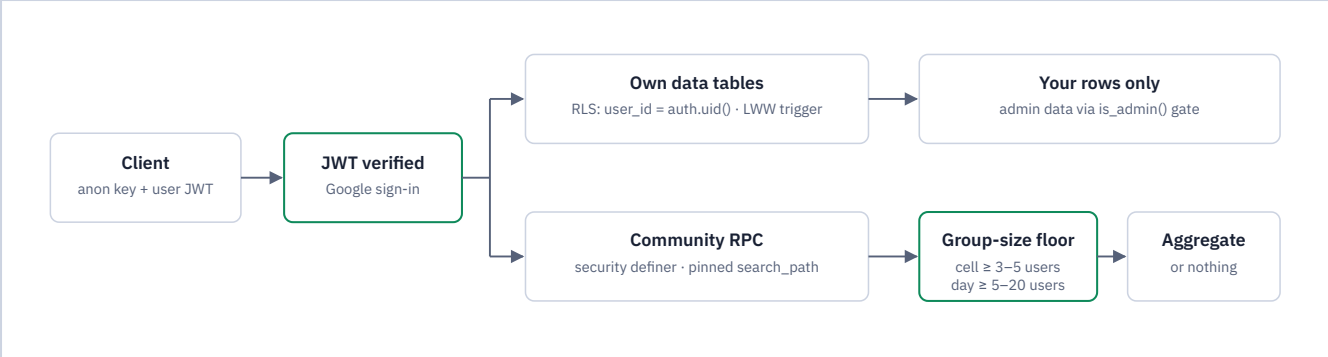


Figure 11. Authorisation path for a request. Cross-user data is only ever an aggregate, and only above a minimum group size.

3.4 Controls

AREA	CONTROL
Consent	The assistant is off by default. Explicit consent with a list of what is and is not sent; withdrawn in one click. Notes are previewed, redacted, before they are sent. Accepting the Terms and Privacy Policy is a separate, required step before sign-in.
Access	Deny by default: `AI_ALLOWED_USERS` lists who may use the function, empty means nobody. JWT verified. E-mail allowlist entries match Google sign-ins only.
Abuse and cost	Per-user daily cap counted in the database before the model is called; a failing counter stops the call. Every agent step counts. 90 KB request limit. Notes cached by hash and sent once.
Prompt injection	System prompts live only on the server. Question and results travel inside JSON strings labelled as data. Tool names are validated against the catalogue; no tool can write. Replies are validated, markup is stripped and the page renders text only.
Hallucination	Answers are built from results; every number is checked against them and marked when unmatched; the results are shown under each answer.
Explainability	Each recommendation shows its rule, setting, out-of-sample and in-sample result, sample sizes and ranges.
Auditability	"What was sent" shows every request, tool run with its tier, and answer, with exact bodies, downloadable. The server keeps no content.
Retention	Server: one counter per user and day. Provider: `data_collection: "deny"`, so only endpoints that do not store or train on prompts are used. Device: log of the last 80 entries, notes cache, rules, paper decisions, all deletable.

AREA	CONTROL
Client hardening	MV3 default CSP, no remote code, no `externally_connectable`. Dashboard CSP `script-src 'self'; object-src 'none'; base-uri 'none'`. Closed shadow root for the in-page card, which ignores synthetic events. Every HTML sink fed by imports, other users or AI replies is escaped; CSV exports neutralise formulas.
Supply chain and CI	Third-party GitHub Actions pinned to commit SHAs; Pages publishes only web-app files; PR previews deploy only after a maintainer label.
Kill switch	Remove a user from the allowlist or unset the key; the rest of the app is unaffected. Provider error text is never returned to the client.

3.5 Security review and known risks

A read-only code review on 17 October 2026 covered database authorisation, edge functions, the extension, and build and hosting. Each reported issue was re-checked against the code by a second, independent reviewer. It confirmed three high-severity findings and a set of medium and low ones. All three high findings were fixed or partly fixed in code, with the remaining steps assigned as owner actions (for example, moving store-publishing secrets to a protected environment).

ID	HIGH FINDING	FIX	STATUS
H1	A crowd-overlap function had no minimum group size and returned precise values, from which other users' views could be reconstructed	20-user day and 5-user cell floors; anon access revoked	fixed
H2	User feedback text could steer an internal automation agent holding a repository token	Feedback insert locked down; the agent's shell limited; feedback marked untrusted	partly fixed
H3	The public site published the whole private repository	Pages deploy limited to web-app files	fixed

The main residual risks of the AI layer, from the risk register:

#	RISK	MITIGATION	RESIDUAL
R1	The model states a wrong number or explains a true one wrongly	Numbers checked and marked; results shown	A wrong explanation of a right number is not detected

#	RISK	MITIGATION	RESIDUAL
R2	Personal data typed into a question or a note	Redaction of obvious patterns; preview; consent text	Names in free text cannot be reliably recognised
R3	Re-identification from the AI agent's per-decision records	Aliases, day offsets, request-local ids	A price path matched to public prices can date a record
R4	A model endpoint stores or trains on what it is sent	`data_collection: "deny"` and `require_parameters` on every OpenRouter request; endpoints that may train are never routed to	Relies on the policies endpoint providers declare to OpenRouter
R7	A rule looks good by luck	Walk-forward with embargo; 99% range; 0 of 216 in null worlds	Overlapping holding periods make ranges somewhat optimistic
R14	Hindsight in the AI agent's backtest	Asked only about moments ≤ 2 days old; saved once; no backfill	Few AI decisions at first
R15	Paper decisions read as advice	Labelled as paper portfolios; acts only on the user's own buys and looks; no order path	A user may still copy a "buy"

3.6 Regulatory compliance: EU, China, United States

Three design choices carry most of the compliance load in every jurisdiction: trades and account data never leave the browser, the product describes the investor's own behaviour instead of recommending securities, and the model is a replaceable explainer behind a deny-by-default gate. The tables show how each rule is met today and what a launch in that market would add. The one feature that needs a market-by-market decision is the AI agent's stock-level paper verdict, which is the closest the product comes to investment advice.

European Union (current market)

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
GDPR	Lawful basis, purpose limitation, minimisation, transparency, user rights	Consent before any transfer; trades never leave the device; "What was sent" log; delete all; EU hosting	Data processing agreement with the model provider; record of processing

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
MiFID II	Personal recommendations on financial instruments are investment advice and need authorisation	Describes the investor's own behaviour and evidence; no forecasts, no orders, no securities recommendations	Legal review of the AI agent's stock-level paper verdicts

People's Republic of China

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
Personal Information Protection Law (PIPL), Art. 3 and 53	Applies to processing outside China that serves or analyses the behaviour of individuals in China; a representative in China is required	Same design as for GDPR: minimisation, consent, local storage	Appoint a representative in China; Chinese-language notices
PIPL Art. 28 and 29	Financial account information is sensitive personal information and needs separate consent	Imported trades and account data are processed only on the device and never collected	Separate consent screens for each feature that sends data
PIPL Art. 38 and 39; 2024 cross-border data flow provisions; certification measures in force since 1 Jan 2026	Separate consent for transfers abroad. Below 100,000 individuals a year (non-sensitive data) no CAC review is needed; above that, standard contract or certification	Only attention events and de-identified results leave the device; no sensitive data is transferred	Track user counts; host Chinese users' data in China if they grow beyond the exemption

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
Interim Measures for Generative AI Services (2023)	Services offered to the public in China need security assessment and algorithm filing where they have public-opinion attributes; foreign services can be restricted	The model is replaceable through one module; it only explains results computed on the device	Switch the provider module to a model filed in China
Measures for Labelling AI-Generated Content (in force 1 Sep 2025)	Explicit and implicit labels on AI-generated text	Every assistant answer is shown as the assistant's, with the results it was built from	Add the prescribed explicit label and metadata label
Securities investment consulting licence (CSRC)	Analysis, forecasts or advice on securities offered to investors need a CSRC licence	No forecasts and no securities recommendations; rules concern timing, size and research habits	Switch off stock-level AI agent verdicts, or partner with a licensed firm

United States

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
Investment Advisers Act of 1940	Advice on the value of securities for pay requires registration; the publisher's exclusion (*Lowe v. SEC*, 1985) covers only impersonal, general publications	Behavioural analytics on the investor's own trades; no advice on buying or selling a security	Legal review of the AI agent's stock-level paper verdicts. The SEC withdrew its predictive data analytics proposal on 12 June 2025
FTC Act, section 5	No unfair or deceptive practices, including claims about AI	Claims rest on out-of-sample evidence with ranges; no promised returns; "not enough data" is shown when true	Review marketing copy against the same rule

RULE	WHAT IT REQUIRES	HOW STOCKTRAILS MEETS IT	STILL TO DO
Gramm-Leach-Bliley Act, FTC Safeguards Rule	Financial institutions must protect customer financial information	The developer never receives account or trade data; it stays in the browser	Re-assess if trade data is ever synced
State privacy laws (California CCPA/CPRA and others)	Notice, access, deletion and opt-out of sale, above revenue or volume thresholds	Access through export, deletion through "Delete all", no sale of data	Add state-specific notices once a threshold is reached
Colorado SB 26-189 (signed 14 May 2026, in force 1 Jan 2027)	Notices and human review for automated decision tools that materially influence consequential decisions such as lending	The app makes no decisions about consumers; every change is the investor's click	None expected; monitor the rulemaking
DOJ rule on bulk sensitive data (28 CFR Part 202, in force 8 Apr 2025)	Limits giving countries of concern, including China, access to personal financial data of more than 10,000 US persons	No personal financial data is collected centrally	Keep US users' data away from any vendor or partner controlled from China

This is a design assessment, not legal advice. Model risk is contained the same way everywhere: the statistics that drive recommendations contain no model, and the model's influence is limited to choosing read-only tools, wording, and paper decisions.

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